

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
```

```
In [2]: iris = load_iris()
```

```
In [3]: df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species'] = df['species'].map(dict(enumerate(iris.target_names)))
df.head()
```

```
Out[3]:
```

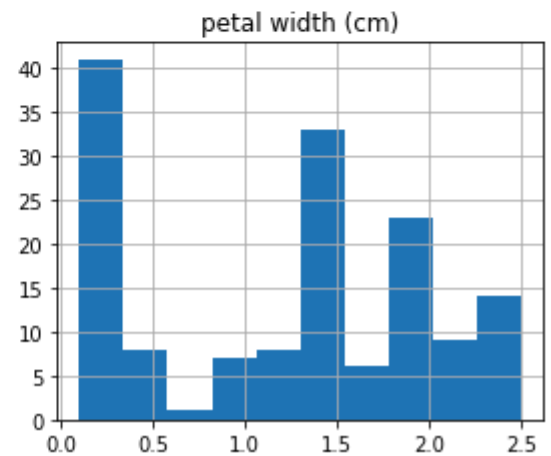
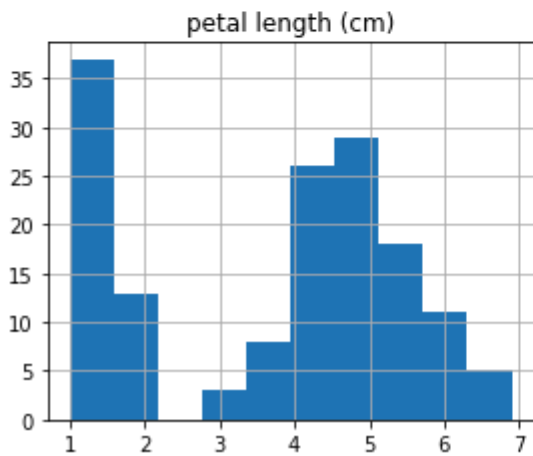
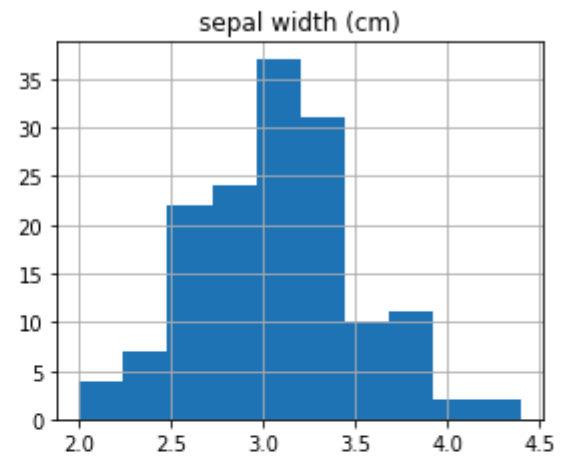
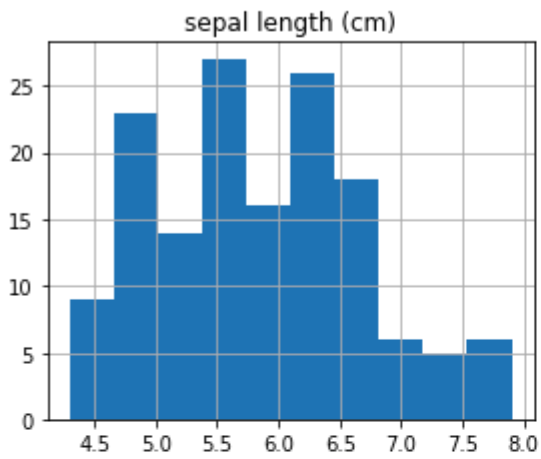
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [4]: df.info()
```

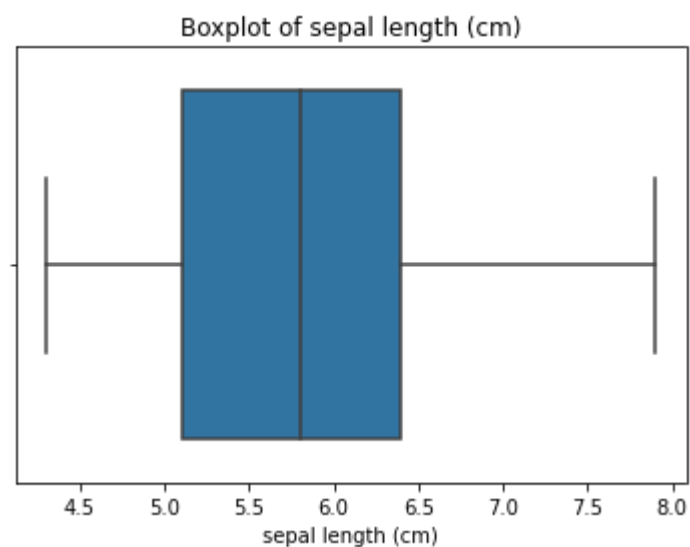
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   sepal length (cm)      150 non-null   float64
1   sepal width (cm)       150 non-null   float64
2   petal length (cm)      150 non-null   float64
3   petal width (cm)       150 non-null   float64
4   species                150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

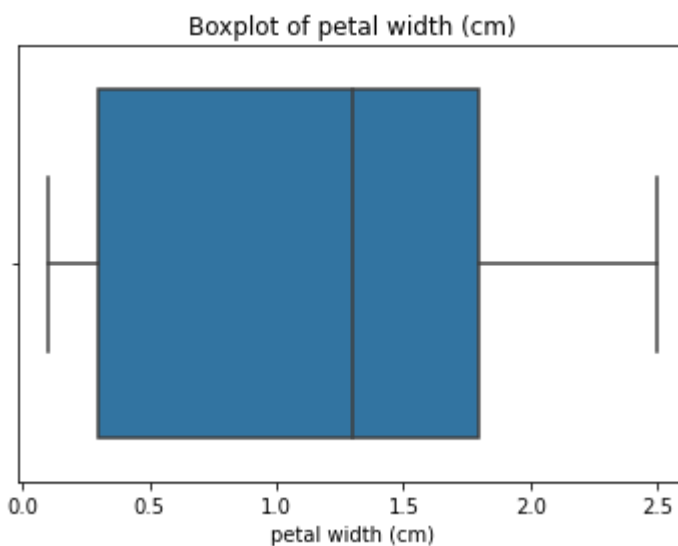
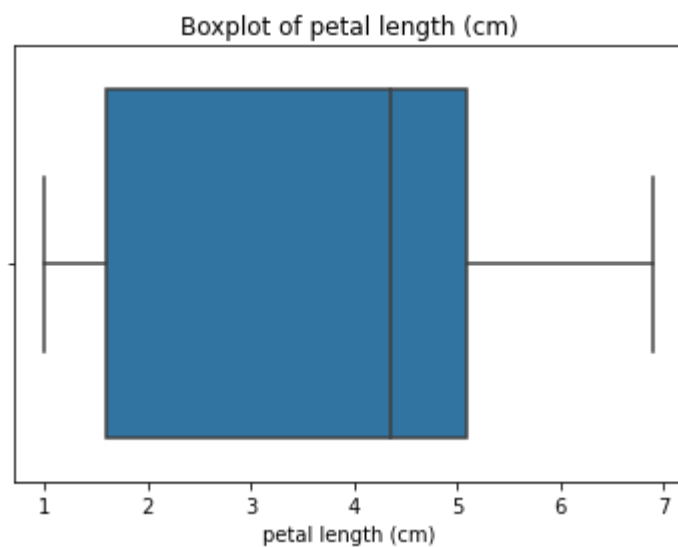
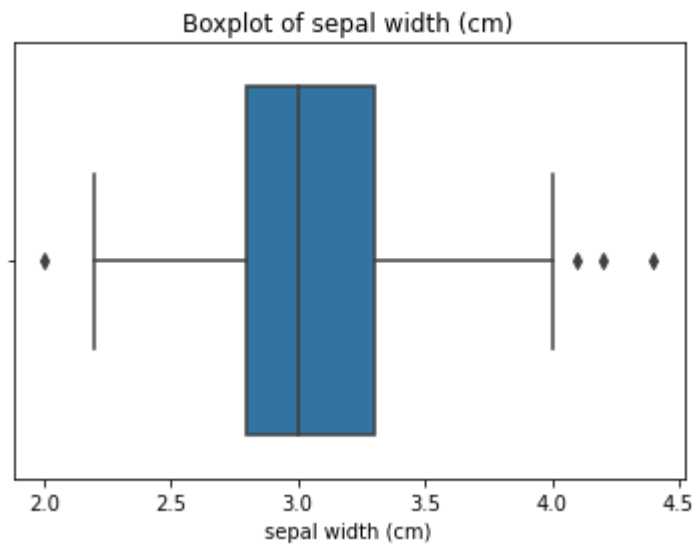
```
In [5]: df.hist(figsize=(10,8))
plt.suptitle("Histograms of Iris Features")
plt.show()
```

Histograms of Iris Features

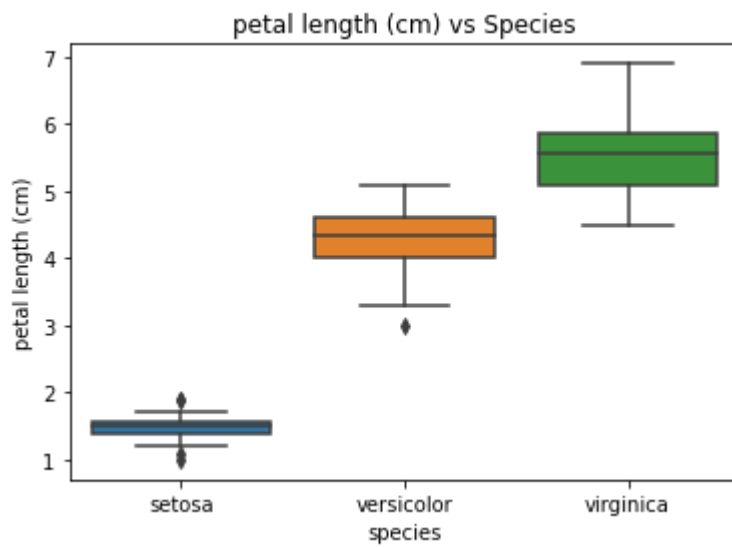
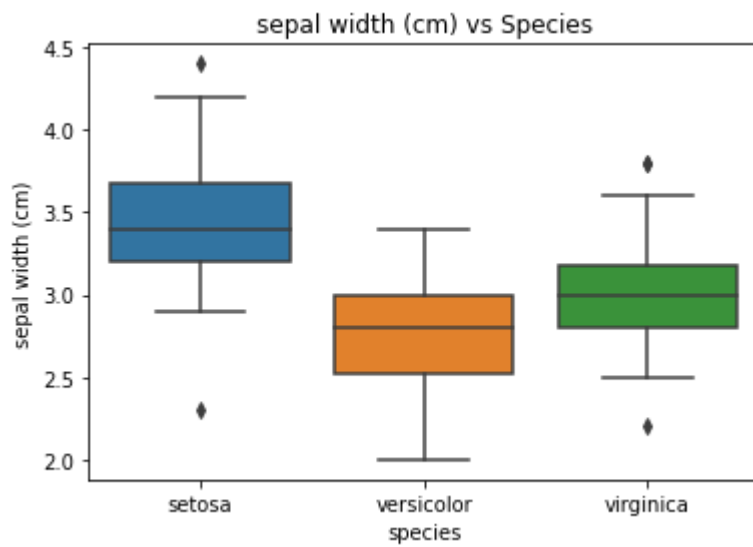
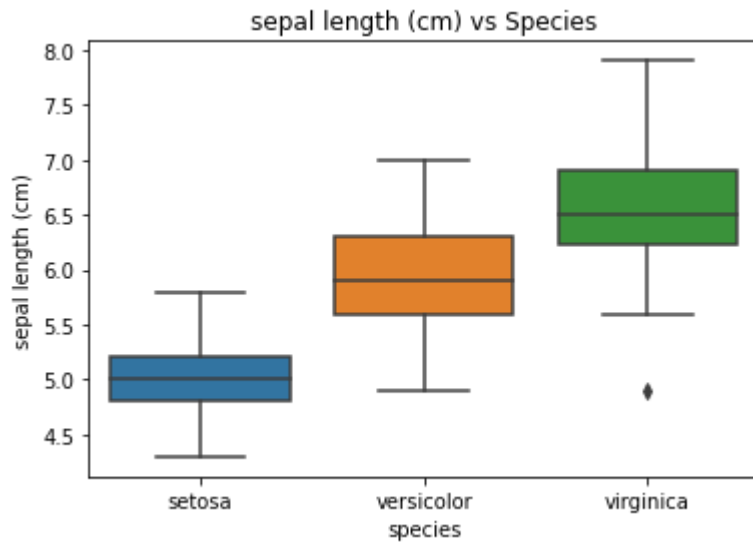


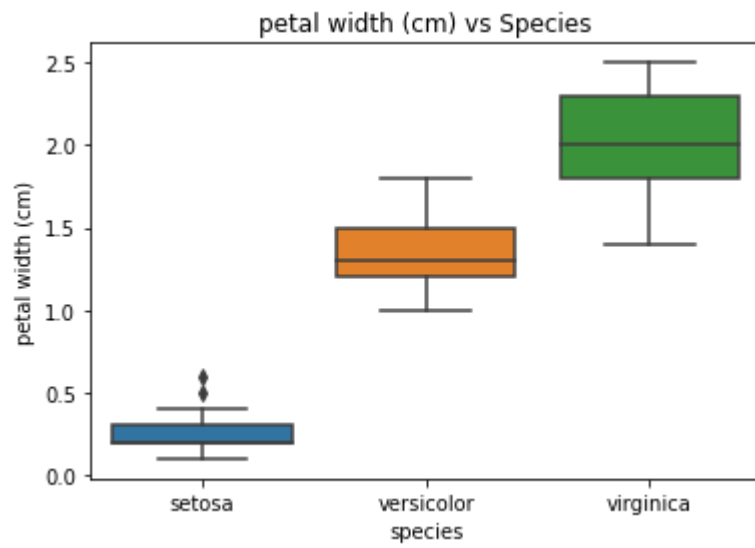
```
In [6]: for col in df.columns[:-1]:
        plt.figure()
        sns.boxplot(x=df[col])
        plt.title(f"Boxplot of {col}")
        plt.show()
```





```
In [7]: for col in df.columns[:-1]:  
        plt.figure()  
        sns.boxplot(x='species', y=col, data=df)  
        plt.title(f"{col} vs Species")  
        plt.show()
```





In []: